INCH POUND

MIL-P-70612A (AR)
02 August 1993
SUPERSEDING
MIL-P-70612 (AR)
6 August 1986

MILITARY SPECIFICATION

PROJECTILE, 155MM: M864 BASE ASSEMBLY FOR

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality evaluation provisions, and packaging for Projectile, 155MM: M864 Base Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1320
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-M-6867	- Magnetic Inspection Units
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1949	- Inspection Process, Magnetic Particle

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Orders Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS

9381119	- Base Assembly

PACKAGING DRAWINGS

12561471	- Packing & Marking for Base Assembly Metal Part for M864 155mm Projectile

INSPECTION EQUIPMENT DRAWINGS

9280375	- Base Magnetic Particle Standard for 155MM, M864
9280376	- Base Closure Magnetic Particle Standard for 155MM, M864

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Method E-8	- Tension Testing of Metallic Materials
ASTM Method E-10	- Brinell Hardness of Metallic Materials
ASTM Method E-103	- Rapid Indention Hardness Testing of Metallic Materials
ASTM Method B-117	- Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.2 Base and base closure heat treatment.

3.2.1 Heat treatment. Each component shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering. In accordance with accepted machining procedures, the component configuration prior to heat treatment shall minimize stress risers, sharp corners, and grooves with sharp roots in order to minimize the possibility of quench cracking, as well as contain sufficient material allowance to allow for removal of all decarburization, if any, during post heat treatment machining operations. The contractor shall submit both the heat treatment and the forging billet heating process control documents to the technical agency for approval.

3.3 Metal defects.

3.3.1 Components or assemblies. All components or assemblies shall be free from all metal defects, in accordance with the defect standards (see 3.3.2) including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits.

3.3.2 Defect standards. Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first month's production or whenever requested by the procuring contracting agency.

3.4 Proving ground function test. The base assembly metal parts shall reveal no evidence of failure when tested. Metal parts failure shall be loss or breakup of metal part component(s) in flight. When fired for ejection, base assembly and cargo not ejecting shall be cause for rejection (the failure analysis must determine the cause to be the base assembly).

3.5 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.6. Control dimensions.

3.6.1 Tool control dimensions. Dimensions marked tool control shall be gaged after each shift of production on each machine in operation and prior to any tooling change. When destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted, provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance and instituting corrective action that assures drawing requirements are being met throughout the life of the tool.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

3.6.2 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to Level III sampling plan (see 4.5.2a).

3.7 Protective finish. Details of the proposed procedure for zinc phosphate coating that is to be used by the contractor shall be submitted in writing to the Government Technical Agency, and written approval shall be received prior to initiation of production. The proposed procedure shall include a description of the materials, process, equipment to be used by the contractor, as well as a detailed method of control including limits for time, temperature, concentration and all other information pertinent to the process. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, and referenced specifications.

3.8.1 Burr and sharp edges. No component shall have a burr or sharp edge which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.8.2 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease, oil, rust, or other foreign matter which may interfere with final inspection and completely free of same prior to shipping.

3.8.3 Threads. Threads shall be full and undamaged, for the entire minimum length and/or depth, as required by the applicable drawings.

3.8.4 Painting. Painting shall be complete, void free, and undamaged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) and for the provisions of MIL-A-48078 as specified herein. Except as otherwise specified in the contract or purchase order, the contract may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (See 4.3)

b. Quality Conformance Inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Base closure	9381120	7
Base	9381122	7
Base assembly	9381119	22

Twenty of the above base assemblies shall be forwarded to a designated load plant for loading and shall then be forwarded to a Government proving ground for ballistic testing.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE First Article Inspection of Base Assembly for Projectile 155MM, M864	SHEET 1 OF 1		DRAWING NUMBER See below NEXT HIGHER ASSEMBLY Not applicable
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Base Closure (Drawing 9381120)			
	Examination for defects	5 units	3.1	4.4.2.2
	Yield strength and percent elongation	4 specimens	3.1	4.5.1.2
	Base (Drawing 9381122)			
	Examinations for defects	5 units	3.1	4.4.2.1
	Yield strength and percent elongation	6 specimens	3.1	4.5.1.2
	Base Assembly (Drawing 9381119)			
	Examination for defects	22	3.1	4.4.2.3
	Salt spray test	2 specimens	3.1	4.5.3
	Ballistic test	-	3.4	4.5.4
	a. Excess pressure	10	3.4	4.5.4.1
	b. Zone seven	5	3.4	4.5.4.2
	c. Service pressure	5	3.4	4.5.4.3

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of base assemblies shall contain bases and base closures from not more than one manufacturer.

4.4.1.1 Component heat treatment lot. A heat treatment lot of any component shall consist of components from only one mill heat of steel that have been heat treated in the same heat treatment equipment at the same temperature and for the same time cycles, in one unchanged process. The maximum lot size shall be 2000 components. If a mill heat exceeds 2000 components, the mill heat shall be divided into any convenient number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots.

4.4.1.2 Miscellaneous lot. Components manufactured from small heats of steel, or those that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The maximum lot size shall be 2000 components.

4.4.1.3 Use. The heat treatment lot constituted per 4.4.1.1, 4.4.1.2 shall be the basis for selecting samples for mechanical properties tests. Components in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

4.4.1.4 Base assembly lot. Heat treatment lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into base assembly lots.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics prargraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characterisics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, with Table II using the inspection levels stated in the Classification of Characteristics paragraphs.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE II. Attributes sampling inspection.

LOT SIZE	INSPECTION LEVELS					
	I	II	III	IV	V	VI
2 to 8	*	*	*	5	5	3
9 to 15	*	*	13	5	5	3
16 to 25	*	*	13	5	5	3
26 to 50	*	*	32	5	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	125	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13

Numbers under inspection levels indicate sample size; asterisks indicate one hundred percent inspection. If the sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation. (see 6.11). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alterative(s0 provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

b. Automatic gaging,. Facilities equipped with government furnished automatic gaging capable of performing 100 percent inspection of any characteristic shall utilize this capability in accordance with the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Quality conformance inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Base	1 OF 3	9381122
			NEXT HIGHER ASSEMBLY: 9381119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Yield strength and percent elongation	note 1	3.1	4.5.1.2
b	Metal defective (see 6.12)	100%,0-1	3.3.1	4.5.2.1
Major				
101	Pitch diameter of closure threads	Level III	3.1	Gage
102	Major diameter of closure threads	Level III	3.1	Gage
103	Pitch diameter (maximum) of body threads	Level III	3.1	Gage
104	Major diameter of body threads	Level III	3.1	Gage
105	Pitch diameter of external boss threads	Level III	3.1	Gage
106	Major diameter of external boss threads	Level III	3.1	Gage
107	Pitch diameter of internal boss threads	Level III	3.1	Gage
108	Minor diameter of internal boss threads	Level III	3.1	Gage
109	Inside diameter	Level III	3.1	Gage
110	Large shoulder diameter	Level III	3.1	Gage
111	Smaller shoulder diameter	Level III	3.1	Gage
112	Diameter of body thread undercut	Level III	3.1	Gage
113	Distance from forward face to end of body thread undercut	Level III	3.1	Gage
114	Diameter of forward shoulder undersize	Level III	3.1	Gage
115	Internal depth of boss	Level III	3.1	Gage
116	Height of boss	Level III	3.1	Gage
117	Diameter of closure thread relief	Level III	3.1	Gage
118	Diameter of rear closure threads (2 places)	Level III	3.1	Gage
119	Diameter of O-ring groove	Level III	3.1	Gage

NOTES:

1. Sample size in accordance with inspection method paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.1	Base		9381122
			NEXT HIGHER ASSEMBLY
			9381119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
120	Length from shoulder to forward face	Level III	3.1	Gage
121	Length from shoulder to rear face	Level III	3.1	Gage
122	Length from rear face to (rear of) rear O-ring groove	Level III	3.1	Gage
123	Width of rear O-ring groove	Level III	3.1	Gage
124	Length from start of closure threads to shoulder	Level III	3.1	Gage
125	Total length	Level III	3.1	Gage
126	Depth of cavity	Level III	3.1	Gage
127	Projected runout (2 places) of closure thread pitch diameter	Level III	3.1	Gage
128	Minimum length of internal boss thread	Level III	3.1	Gage
129	Minimum length of external boss thread	Level III	3.1	Gage
130	Projected runout (2 places) of closure thread pitch diameter	Level III	3.1	Gage
131	Runout of inside diameter	Level III	3.1	Gage
132	True position of diameter rear of closure threads (2 places)	Level III	3.1	Gage
133	Straightness of forward face	Level III	3.1	Gage
134	Straightness of rear face	Level III	3.1	Gage
135	Width of body thread undercut	Level III	3.1	Gage
136	Width of forward chamfer	Level III	3.1	Gage
137	Width of forward O-ring groove	Level III	3.1	Gage
138	Pitch diameter (minimum) of body threads	Level III	3.1	Gage
139	Distance from shoulder to forward O-ring groove	Level III	3.1	Gage
140	Diameter forward of shoulder (oversize)	Level III	3.1	Ring Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Base	3 OF 3	9381122
			NEXT HIGHER ASSEMBLY 9381119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
141	True position of diameter forward of shoulder	Level III	3.1	Gage
142	Diameter of forward O-ring groove	Level III	3.1	Gage
143	Pitch diameter of body threads	Note 2	3.1	"Variables" Measurement Gage
Minor				
201	Width of closure thread relief	Level V	3.1	Gage
202	Lot identification not present or illegible	Level V	3.1	Visual
203	Protective finish inadequate or damaged	Level V	3.1	Visual
204	Surface finish improper	Level V	3.1	Visual
205	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

2. "Variable" measurements shall be performed in accordance with the contractor's SPC plan (when approved). Functional gages shall be used for lot acceptance. "Variables" gage readings shall be used for process control purposes (see Major Defects 103, 138, and 143).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Quality conformance inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Base Closure		9381120
			NEXT HIGHER ASSEMBLY
			9381119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Yield strength and percent elongation	note 1	3.1	4.5.1.2
b	Metal defective (see 6.12)	100%,0-1	3.3.1	4.5.2.1
Major				
101	Total length	Level III	3.1	Gage
102	Location of forward basic outside diameter	Level III	3.1	Gage
103	Location of rear basic outside diameter	Level III	3.1	Gage
104	Diameter of thread pilot	Level III	3.1	Gage
105	Depth of cavity	Level III	3.1	Gage
106	Thread pitch diameter	Level III	3.1	Gage
107	Thread minor diameter	Level III	3.1	Gage
108	Depth of largest rear cavity basic diameter	Level III	3.1	Gage
109	Web thickness at rear face	Level III	3.1	Gage
110	Diameter of burner hole	Level III	3.1	Gage
111	Forward runout of boat tail to datums	Level III	3.1	Gage
112	Rear runout of boat tail to datums	Level III	3.1	Gage
113	Depth of notches (4 places)	Level III	3.1	Gage
114	Diameter of notches (2 places)	Level III	3.1	Gage
115	Straightness of forward face	Level III	3.1	Gage
116	Wall thickness at basic diameter	Level III	3.1	Gage
117	Distance between basic diameters on exterior aft cavity	Level III	3.1	Gage
118	Depth from forwardmost interior datum to rearmost interior cavity surface	Level III	3.1	Gage

NOTES:
1. Sample size per test method paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Base Closure		9381120
			NEXT HIGHER ASSEMBLY 9381119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
119	Runout of base closure cavity to datums	Level III	3.1	Gage
120	Runout of largest rear cavity diameter to datums	Level III	3.1	Gage
121	Runout of base closure orifice to datums	Level III	3.1	Gage
Minor				
201	Diameter of thread relief (if applicable)	Level V	3.1	Gage
202	Width of thread relief (if applicable)	Level V	3.1	Gage
203	Depth from forward face to threads	Level V	3.1	Gage
204	Diameter of counterbore rear of thread relief	Level V	3.1	Gage
205	Depth to rear of thread relief	Level V	3.1	Gage
206	Lot identification not present or illegible	Level V	3.1	Visual
207	Radius in bottom of notches (4 places)	Level V	3.1	Gage
208	Protective finish inadequate or damaged	Level V	3.1	Visual
209	Surface finish improper	Level V	3.1	Visual
210	Evidence of poor workmanship	Level V	3.8	Visual
211	Effective length of thread, minimum	Level V	3.1	Gage
212	Perfect form of notches, 4 places equally spaced, at maximum material condition	Level V	3.1	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Base Assembly	1 OF 1	9381119
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Ballistic test	Note 1	3.4	4.5.4
b	Body threads damaged (prior to application of protective cap) (see 6.14)	100%,0-1	3.1	Visual
c	Magnetic particle inspection stamp missing on base or base closure indicating a failure to pass test (prior to assembly of thread protector) (see 6.13)	100%,0-1	3.1	Visual
Major				
101	Salt spray test	Note 1	3.1	4.5.3
102	Protective coating incomplete or damaged	Level III	3.8.4	Visual
Minor				
201	Protective cap missing	Level V	3.1	Visual
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:
1. Sampling per inspection method paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Quality conformance inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.4	Packing & Marking for Base Assembly Metal Part for M864 Projectile	SHEET 1 OF 1	12561471
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Improperly packed	Level V	3.1	Visual
202	Marking missing, incorrect, or illegible	Level V	3.1	Visual
203	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the Quality Conformance Inspection Tables and herein.

4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410. As a minimum, operating personnel shall be level I certified, and calibrating personnel shall be level II certified.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Hardness of base and base closure. In order to determine uniformity of heat treatment each component (base and base closure) will be subjected to a rapid indentation hardness test (Brinell alternative) after heat treatment. The test shall be made in accordance with ASTM Method E-103. The surface of each component, in the area to be hardness tested, shall be prepared by machining (or grinding) a flat of sufficient depth, 0.015 inches minimum, to remove scale and get below the decarburization zone. The flat shall be prepared in such a manner so that the mechanical properties of the underlying material is unchanged, and shall be of sufficient size and flatness so as to minimize error due to improper registration of the tester with the component. Manual grinding is prohibited. Flats and impressions shall be completely removed by final machining. Periodic verification of the hardness tester shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-103 in lieu of production samples. The tester shall apply a preload followed by application of the full load. The hardness tester shall use a standard 10mm Brinell carbide ball, and the indentor penetration shall be continuously measured, by automatic means, for the peroid beginning with the application of the major load, and ending after the prescribed dwell time for each component tested. The component shall be securely supported so that no rocking or shifting of the component will occur during the test. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. All hardness test measurements shall be automatically displayed and recorded at the conclusion of the test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

4.5.1.1.1 Base hardness test location. Reading shall be taken in the center of the forward face of the base.

4.5.1.1.2 Base closure hardness test location. Readings shall be taken on the outside sidewall one inch forward of the rear face of the base closure.

4.5.1.2 Yield strength and percent elongation. The two hardest and two softest components from each standard heat treatment lot or the four hardest and eight softest components from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.1 shall be subjected to this test. Samples for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. The criteria for performing the tests shall be as prescribed in ASTM Method E-8. Failure of any longitudinal test specimen to comply with the yield strength and elongation requirements, or failure of any transverse test specimen to comply wih the yield strength requirements of the applicable drawing shall be cause for rejection of the heat treatment lot. The transverse elongation results shall be recorded for information only.

At the option of the contractor, the entire lot may be re- heat treated and retested; or a new component may be selected and tested from a segment of the heat treatment lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this treatment lot, components falling within this range will be accepted. All other components will be rejected. Rejected components may be re-heat treated and retested. All tensile specimens will be the largest obtainable round type specimen as prescribed in ASTM Method E-8.

4.5.1.2.1 Base. For each base tested two longitudinal test specimens shall be taken from opposite sides of the base such that the center of the gage length shall be midway of the length of the base and one transverse specimen shall be taken from the web of the base such that the center of the gage length will be in the center of the base at mid web thickness.

4.5.1.2.2 Base closure. For each closure tested two longitudinal specimens shall be taken from opposite sides of the base closure such that the center of the gage length will be one inch from the forward face of the closure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.5.1.3 Test reporting. Test information for the base and base closure shall include:

a. Material and specimen identification, referencing the mill heat of material and heat treat lot.

b. Yield strength and the method used to determine the yield strength per ASTM Method E-8.

c. Tensile strength.

d. Elongation (per ASTM Method E-8).

e. Specimen type per ASTM Method E-8.

f. Speed and method used to determine speed of testing.

g. Type of holder or grips used.

h. Hardness per ASTM Method E-103.

i. Specimen test section dimensions.

j. Reasons for replacement specimens.

k. A copy of the load-deflection curve.

4.5.2 Finished components.

4.5.2.1 Bases and base closures. Each component after final machining shall be inspected by the magnetic particle inspection technique. At the option of the contractor, this test may be performed prior to machining either the threads or notches if applicable. The procedures for performing this test shall be as specified in MIL-STD-1949, for wet fluorescent method with indirect magnetization by the continuous method. The entire component, inside and outside, shall be inspected after inducing a longitudinal magnetic field, and then the part shall be rotated 90 degrees and the longitudinal field re-induced and the component reinspected. To establish inspection sensitivity magnetic particle standard drawing 9280375 shall be used for the base and 9280376 shall be used for the base closure. Any component which fails to comply with the requirements of 3.3 shall be rejected. Components which comply shall be identified with the letter "M" as specified on the applicable drawing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

The solids content of the magnetic particle suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each shift of production (not to exceed 10 hours) and at intervals of each half-shift (not to exceed 5 hours) of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-STD-1949. When performing this test for solids content of the suspension, if it is determined that it does not comply, all components that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867.

4.5.3 Salt spray test. The salt spray test shall be performed on each base assembly utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable detailed specification for the protective finish, except the curing cycle for the base assemblies shall be in accordance with component dwg. Sampling shall be as specified in TT-C-490. When on reduced sampling (twice per week), one sample shall be taken during the first 30 minutes of production and a second taken during the last day of production that week. Failure to comply will be cause for rejection of the production since the previous test.

4.5.4 Proving ground ballistic test. The following tests shall be conducted using cannon tubes having 25 percent minimum remaining life based upon current gun tube condemnation criteria. Smear photographs shall be taken at 10 and 30 meters in front of the gun muzzle to record the condition of the projectile as it exits the tube. Radar equipment shall be used to track the projectile in flight. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements of 3.4. Range, deflection, and velocity shall be recorded for each round fired for information. An attempt will be made to recover the projectiles fired for first article. During production acceptance tests projectiles shall be recovered only if a failure or suspected failure occurs (see 6.9).

4.5.4.1 Excess pressure. The base assemblies submitted for this test shall be further assembled into projectiles that will contain inert cargo; have a base burner assembly with an inert grain, without igniters; a dummy expulsion charge; and a dummy fuze. Each projectile will be fired with a PXR6297A1 propelling charge temperature conditioned to obtain 60-64 thousand PSI chamber pressure and with a M199 cannon tube. The completely assembled projectiles shall be temperature conditioned to 145 - 5 degrees F.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

4.5.4.2 Zone 7. The base assemblies submitted for this test shall be further assembled into projectiles that will contain inert cargo; have a base burner assembly with an inert grain; without igniters; a live expulsion charge; and a live M577 fuze. Each projectile will be fired in a M185 or M199 cannon tube with an M4A2 propelling charge. The complete projectiles and the propelling charges shall be conditioned to -50 + 5 degrees F. The fuze will be set for airburst. Fuze function time will be recorded.

4.5.4.3 Service pressure. The base assemblies submitted for this test will be assembled into projectiles as in 4.5.4.2. Each projectile will be fired in a M199 cannon tube with a M203 propelling charge. The propelling charges and projectiles shall be conditioned to 70 ± 5 degrees F. The fuze will be set for airburst. Fuze function time will be recorded.

4.5.4.4 Sampling procedure. Normal sampling will be conducted until 5 consecutive lots have been accepted. Reduced sampling shall then be instituted. Any failure, while on reduced sampling, mandates return to normal sampling. Sample sizes for normal and reduced sampling shall be as follows:

	Normal	Reduced
a. Excess pressure	5	2
b. Zone seven	3	3
c. Service pressure	2	-

5. PACKAGING

5.1 Packing. This section (Section 5) is not applicable if LAP operations are to be performed at the same facility.

5.2 Packaging, packing, and marking.

5.2.1 Base Assembly. Base assembly packaging, packing and marking shall be in accordance with drawing 12561471.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The base assemblies covered by this specification are intended for the Projectile, 155MM: M864.

6.2 Acquisition requirements. Acquision documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Requirements for submission of first article sample.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract per the Contact Data Requirements List, DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.1 3.6.2 3.7	DI-T-5372	Procedure Report	See 6.3.1

6.3.1 Tailoring. Delete a,b,c,d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits. (Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, VOL. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Submission of inspection equipment designs for approval. Gage designs, as required, shall be submitted to: Commander, ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address shall be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, Data Item OT-90-12009 shall apply.

6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, ARDEC, ATTN: SMCAR-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:29Z
Check the source to verify that this is the current version before use.

6.6 Steel blooms, billets, slabs and bars.

6.6.1 Steel grade. It is the responsibility of the manufacturer of the base assembly to select a steel of a grade(s) compatible with abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machinist to produce the metal parts to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

6.6.2 Raw material. The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade (selected by the base assembly manufacturer) and complying with the specific ASTM designation and tests selected by the base assembly manufacturer.

6.6.3 Forging quality processing techniques. In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machinist have been engineered in terms of equipment, sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

6.7 Preproduction sample/hardness range. To minimize the need for reheat treating material it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening components during processing of the heat treatment lot from that mill heat. When using a hardness range, acceptance will still be in accordance with para. 4.5.1.

6.8 Subject term (keyword) listing.

Forgings, steel, artillery, base

6.9 Ballistic tests. At the option of the Government, ballistic acceptance tests of base assemblies may be conducted in conjunction with acceptance tests of the body and ogive.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

6.10 First Article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample from the first lot of production items, a standard production item from the contractor's current inventory (see 3.5), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instrucions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.11 Submission of alternate quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.12 Metal Defective (Magnetic Particle Inspection). Upon detection of a magnetic particle indication, a visual examination shall be made to determine if the indication is caused by a discontinuity in the surface. If the indication is caused by a surface discontinuity, the part shall be considered to have a critical defect. If the indication cannot be identified as a surface discontinuity, the part shall be considered to have a minor defect and shall be permanently rejected.

6.13 Magnetic particle inspection stamp missing. In the event a metal defective stamp is discovered to be missing, the part shall be removed from the assembly and inspected per its appropriate paragraph for all defects. If that part exhibits any indication, that part shall be considered critically defective, since it evidences a critical breakdown in the inspection and/or material handling system. If the part shows no indication, that part shall be classified as having a minor defect and may be made acceptable by application of the appropriate metal inspection stamp. Non-affected components may be returned to production.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

6.14 Body threads damaged. If the damage to the body threads is cosmetic in nature, the part shall be considered to have a minor defect, but shall be permanently rejected. Damage which results in loss of thread metal shall be considered a critical defect.

6.15 Changes from previous issue. Marginal notations are not used in this specification to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-AD06)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70612A (AR)	930802

3. DOCUMENT TITLE: PROJECTILE, 155MM: M864, BASE ASSEMBLY FOR

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.